

1. Administrative & Study Identification

Full Study Title: A Phase 2, Double-blind, Randomized, Placebo-controlled, Multicenter Study to Evaluate the Efficacy, Safety and Pharmacokinetics of Luspatercept (ACE-536) in Chinese Adult Subjects Who Require Regular Red Blood Cell Transfusions Due to Beta (β)-Thalassemia **Short Title/Acronym:** CA056-001 **Protocol Number:** CA056-001 **NCT Number:** NCT05567458 **Sponsor Name:** Bristol-Myers Squibb **Investigational Product:** Luspatercept (ACE-536) **Phase of Development:** Phase II **Medical Monitor:** Bristol-Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the efficacy of luspatercept plus best supportive care (BSC) versus placebo plus BSC in Chinese adult participants who require regular red blood cell (RBC) transfusions due to β -thalassemia. **Secondary Objectives:** To evaluate the safety, tolerability, pharmacokinetics, and additional efficacy parameters (e.g., higher thresholds of RBC transfusion reduction and transfusion independence) of luspatercept.

2.2 Background & Rationale Beta-thalassemia is an inherited blood disorder characterized by reduced production of hemoglobin, leading to chronic, severe anemia. Patients often depend on regular RBC transfusions to survive, which can result in life-threatening complications such as iron overload. Luspatercept acts as an erythroid maturation agent to reduce the RBC transfusion burden. This trial aims to establish the clinical profile of luspatercept specifically within the Chinese adult population suffering from transfusion-dependent β -thalassemia.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING)

3.2 Planned Enrollment & Duration **Target N\$:** 94 evaluable patients **Number of Sites:** Multicenter study in China **Estimated Study Start:** 16-Oct-2022 **Estimated Primary Completion:** 31-Jul-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant has a documented diagnosis of β -thalassemia or Hemoglobin E/ β -thalassemia (β -thalassemia with mutation and/or multiplication of alpha [α] globin is allowed). 2. Participant is regularly transfused, defined as: 6-25 RBC units in the 24 weeks prior to randomization and no transfusion-free period for > 42 days during that period. 3. Participant has an Eastern Cooperative Oncology Group (ECOG) score of 0 or 1. 4. Participant is willing and able to adhere to the study visit schedule (e.g., not scheduled to receive hematopoietic stem cell transplantation [HSCT]) and other protocol requirements.

4.2 Exclusion Criteria 1. Participant has a diagnosis of Hemoglobin S/ β -thalassemia or α -thalassemia (for example, Hemoglobin H). 2. Participant has an active hepatitis C virus (HCV) infection as demonstrated by a positive HCV ribonucleic acid (RNA) test of sufficient sensitivity, or active infectious hepatitis B virus (HBV) as demonstrated by the presence of hepatitis B surface antigen (HBsAg) and/or HBV deoxyribonucleic acid (DNA) positive, or a known positive human immunodeficiency virus (HIV). 3. Participant has a history of deep venous thrombosis or stroke or thromboembolic events (venous or arterial) requiring medical intervention ≤ 24 weeks prior to randomization. 4. Participant uses chronic anticoagulant therapy, unless the treatment stopped at least 28 days prior to randomization. Anticoagulant therapies used for prophylaxis for surgery or high-risk procedures as well as low-molecular-weight heparin for superficial venous thrombosis and chronic aspirin are allowed. 5. Participant who has extramedullary hematopoiesis (EMH) complications requiring treatment to control the growth of EMH mass(es) during the screening period. 6. Participant used immunomodulatory imide drugs (IMiDs) ≤ 24 weeks prior to randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Luspatercept (Trade Name: Reblozyl, ATC Code: B03XA06) at a recommended starting dose of 1 mg/kg , administered once every 3 weeks. Dose adjustments or titration may occur based on individual hematologic response. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Initial double-blind treatment period of at least 24 weeks followed by longer-term exposure to evaluate sustained safety and efficacy.

5.2 Concomitant & Prohibited Therapy Permitted: Best supportive care (BSC) including routine RBC transfusions and iron chelation therapy. Anticoagulant therapies used for prophylaxis for surgery, low-molecular-weight heparin for superficial venous thrombosis, and chronic aspirin are allowed. **Prohibited:**

Immunomodulatory imide drugs (IMiDs) ≤ 24 weeks prior to randomization; chronic anticoagulant therapy unless stopped 28 days prior to enrollment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Days -35 to 0	Informed Consent, Medical & Transfusion History, HCV/HBV/HIV testing, Baseline Labs
Baseline	Day 0	Randomization, First SC Injection, PK blood draws
Interim	Every 3 Weeks	Administration of Luspatercept/Placebo, safety labs, pre-dose hemoglobin (Hgb) assessment, transfusion tracking
End of Study	Follow-up	Final Vital Signs, Exit Interview, Safety and Thromboembolic event monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants with ≥ 33% reduction from baseline in red blood cell (RBC) transfusion burden over any consecutive 24 weeks. **Measurement Method:** Clinical review of patient transfusion logs and clinical records.

7.2 Secondary & Exploratory Endpoints

- Proportion of subjects with ≥ 33% reduction from baseline in RBC transfusion burden during any rolling 24-week interval compared to the 24-week interval prior to start of investigational product (IP) for luspatercept plus BSC versus placebo plus BSC.
- Proportion of participants with ≥ 33% reduction from baseline in RBC transfusion burden over any consecutive 12 weeks.
- Proportion of participants with ≥ 50% reduction from baseline in RBC transfusion burden over any consecutive 12 weeks.
- Safety endpoints, including the incidence of adverse events (AEs) and thromboembolic events.
- Pharmacokinetics (PK) of luspatercept (e.g., C_{max} , AUC).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via HCP assessment, laboratory abnormalities, and patient reporting at each 3-week visit. Special monitoring for thromboembolic events (TEE) and hypertension. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent safety review committee to evaluate ongoing unblinded safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes. Safety Set for all participants receiving at least one dose of the study drug. **Sample Size Justification:** Target $N = 94$ patients designed to provide adequate statistical power to detect a clinically meaningful difference in transfusion burden reduction between the luspatercept and placebo arms. **Handling of Missing Data:** Categorical efficacy endpoints (like transfusion reduction responder rates) will treat patients with missing data or early dropouts as non-responders. Multiple Imputation (MI) or Mixed-effects Model for Repeated Measures (MMRM) may be utilized for continuous endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard onsite and remote monitoring procedures, verifying source data for transfusion records and AE logs. **Audit Trail:** Secure, 21 CFR Part 11 compliant Electronic Data Capture (EDC) systems with full audit trails for eCRF locking.

11. Study Identifiers & Governance

Posting ID: 806364686057352666 **Primary ID:** CA056-001 **Registry Number:** NCT05567458 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** A Study to Evaluate Luspatercept (ACE-536) in Chinese Participants Who Require Regular Red Blood Cell Transfusions Due to Beta (β)-Thalassemia. **Official Regulatory Title:** A Phase 2, Double-blind, Randomized, Placebo-controlled, Multicenter Study to Evaluate the Efficacy, Safety and Pharmacokinetics of Luspatercept (ACE-536) in Chinese Adult Subjects Who Require Regular Red Blood Cell Transfusions Due to Beta (β)-Thalassemia

12. Clinical Framework (Beta-Thalassemia Specific)

Primary Condition: Transfusion-dependent Beta (β)-Thalassemia **Diagnosis Threshold:** Confirmed via genetic or hemoglobinopathy diagnostics (Hemoglobin E/ β -thalassemia allowed) **Medical Methodology:** Best Supportive Care (BSC) integrated with Erythroid Maturation Agent **Optimization Target:** $\geq 33\%$ reduction in RBC transfusion burden

13. Study Procedures & Methodology

Management Pathway: Integration of luspatercept SC administration with routine transfusion workflows. **Education Protocol (HCP):** Training on proper dosing, reconstitution of luspatercept, and predose Hgb monitoring. **Education Protocol (Patient):** Informed consent process detailing the requirement of visits every 3 weeks and reporting thromboembolic symptoms. **Quality Audit Mechanism:** Clinical monitoring of transfusion logs ensuring strict adherence to the $> 42\%$ transfusion-free exclusion parameters.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 4 years (Start: Oct 2022 to Primary Completion: Jul 2026) **Onsite Visit Cadence:** Once every 3 weeks for treatment administration and monitoring **Remote Monitoring Frequency:** Intermittent safety follow-up calls per study schedule **Checklist Review Interval:** Continuous tracking of RBC units transfused over rolling 24-week periods

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				